5. Block World Planning Using BFS

Python practical code

```
start = (('A', 'B'), ('C',), ())
goal = (('A',), ('C', 'B'), ())

queue = [(start, [])]
visited = [start]

while queue:
    state, path = queue.pop(0)

    if state == goal:
        print(path)
        break

    for i in range(3):
        if state[i]:
            block = state[i][-1]

            for j in range(3):
                if i != j:
                    new = list(state)
                    new[i] = state[i][:-1]
                    new[j] += (block,)
                    new = tuple(new)

                    if new not in visited:
                        visited.append(new)
                        queue.append((new, path + [block]))
```